

Python Workshop

From Python Fundamentals to a Working Application

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Day 1

What We Will Build

A small **Student Management System**

- ▶ **Add** students
- ▶ **View & Find** students
- ▶ **Update & Delete** students
- ▶ **Save & Load** student data

Python concepts $\longrightarrow$ **Application**

The Learning Journey

Python Concepts → **Application Design** → **Working Application**

- ▶ **Functions** – organize reusable logic
- ▶ **Modules** – organize code into files
- ▶ **Exceptions** – handle unexpected situations
- ▶ **Files & JSON** – store and retrieve data
- ▶ **OOP** – model real-world entities
- ▶ **Application Design** – bring everything together

Goal: Understand not just *what* Python can do, but *how* to use it to build an application.

Functions — Why Do We Need Them?

Repeated Code → More Changes → Harder to Maintain

- ▶ Break a large problem into smaller pieces
- ▶ Reuse logic instead of repeating code
- ▶ Give a meaningful name to a piece of logic
- ▶ Make code easier to test and maintain

Problem → Function → Reusable Logic

Modules — Why Do We Need Them?

One Large File → **Multiple Modules** → **Organized Application**

- ▶ Split a large program into smaller files
- ▶ Group related functionality together
- ▶ Reuse code across different parts of an application
- ▶ Make the project easier to understand and maintain

Functions organize logic → **Modules organize code**

Exceptions — When Things Go Wrong

Program → **Unexpected Input** → **Exception**

- ▶ Programs can encounter unexpected situations
- ▶ User input may not be in the expected format
- ▶ Files may not exist
- ▶ External operations may fail

try → **Handle the problem** → **Continue safely**

Files & JSON — Making Data Persist

Program → Data in Memory → Program Ends

What happens to our data?

- ▶ Files allow data to survive after the program ends
- ▶ JSON provides a simple format for structured data
- ▶ Python can write data to JSON files
- ▶ Python can read the data back when needed

Python Objects ↔ JSON File

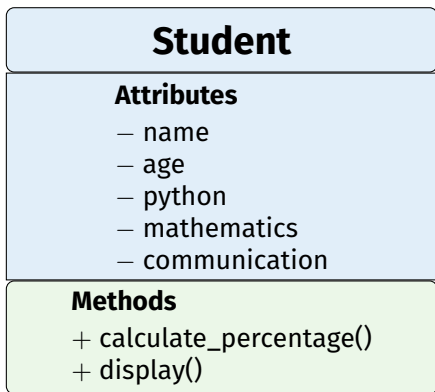
From JSON to an Application

JSON → Data → Application Logic

- ▶ JSON is useful for storing structured data
- ▶ But an application needs more than stored data
- ▶ We need operations such as Add, Find, Update and Delete
- ▶ We need a way to represent real-world entities

Data alone is not an application

OOP — Classes & Objects

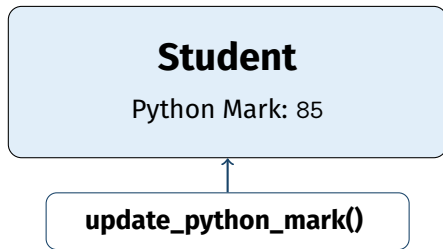


- ▶ A **class** defines the structure and behavior
- ▶ An **object** is an instance of a class
- ▶ Objects combine data (attributes) and operations (methods)
- ▶ This helps us model real-world entities in code

Class → Blueprint

Object → Actual Student

Encapsulation — Controlling Access to Data



- ▶ Protect the object's internal data
- ▶ Control how data is changed
- ▶ Put validation rules in one place
- ▶ Prevent invalid state where possible

Encapsulation → Control how an object's data is accessed or changed

Inheritance — Reusing Common Behavior



- ▶ Put common attributes and behavior in a parent class
- ▶ Child classes inherit that common functionality
- ▶ Child classes can add their own attributes and behavior

Person → **Student / Teacher**

From Concepts to Application Design

Functions

Modules

OOP

JSON / Storage

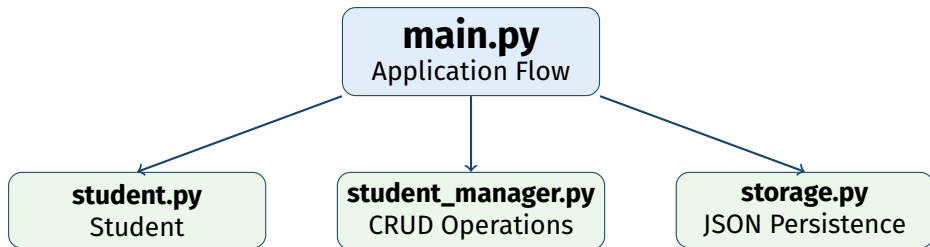


Student Management System

- ▶ Functions organize reusable logic
- ▶ Modules organize the code
- ▶ OOP models the student and related entities
- ▶ JSON provides persistence

Individual concepts become building blocks of an application.

Student Management System — Architecture



Each component has a clear responsibility.

CRUD Operations — Managing Students

Create

Add Student

Read

View / Find

Update

Modify Student

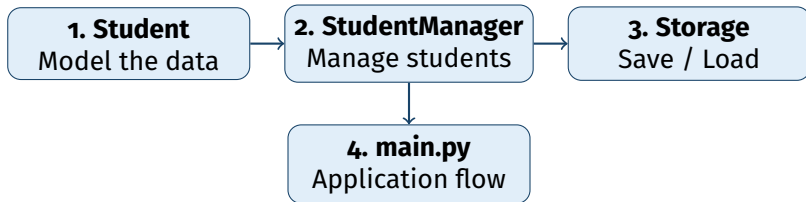
Delete

Remove Student

CRUD → The basic operations needed to manage application data

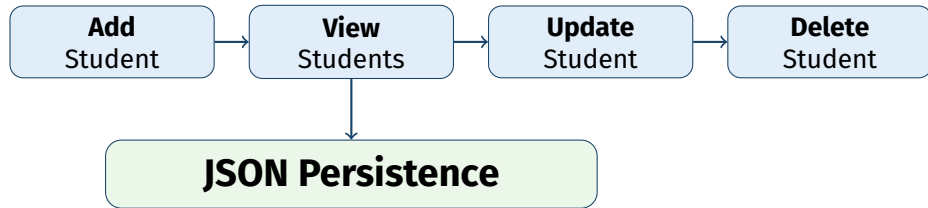
StudentManager provides these operations for our application.

Building the Application



Build one component at a time, then connect them.

Final Demonstration



From Python concepts to a working application

Day 1 — What We Learned

Functions

Modules

Exceptions

Files & JSON

OOP

Application Design

- ▶ We moved from individual Python concepts to an application
- ▶ We used OOP to model real-world entities
- ▶ We separated responsibilities using modules
- ▶ We implemented CRUD operations and JSON persistence

Python concepts → Working Application